

FINLEDGER · ACCOUNTS PAYABLE AUTOMATION

Person 4 — Tally connector (first ERP)

FinLedger AP · India · Tally-first · September 2026 Seat: local agent on :9000, masters sync, GSTIN↔ledger mapping table, idempotent post, attach PDF, outstanding pull.

Product (read this first)

FinLedger is a per-client Accounts Payable machine for Indian CAs and SMBs. Vendors send bills to one official link or one official email. The platform extracts the invoice, scores risk, maps ledgers with client-scoped RAG, waits for a human when the bill is risky, then posts an approved VoucherDraft into that client's ERP. Tally Prime first.

What it does. Takes unstructured vendor PDFs/photos and turns them into posted purchase vouchers with open bills that age. GST / TDS / MSME native. Not a GST filer, not a bank, not a card issuer.

How a bill moves. Vendor sends → Person 2 stores markdown/images → Person 1 extracts CanonicalInvoice and maps a VoucherDraft via RAG → Person 3 scores and reviews → Person 4 posts to Tally. Extraction never posts. The mapping LLM never emits Tally XML.

Full story. docs/PRODUCT.md and docs/HOW_IT_WORKS.md.

Mission

Translate an approved VoucherDraft into Tally Prime XML. Nothing else writes to Tally. Do not start Zoho / QBO / SAP until this loop is boring.

One-line path Person 4 owns:

approved VoucherDraft → validate masters → XML voucher + bill-wise → erp_id → attach PDF
→ outstanding

Layers owned

#	Layer	Person 4 piece
7	Connectors	Router + Tally adapter first. Shared contract listed below.
14.1	Tally Prime	Desktop / on-prem agent → XML HTTP :9000
14.5	Master-data mapping	GSTIN → Tally ledger name. Purchase ledger → Tally name. Tax → input CGST/SGST/IGST.
14.7	Post flow	Idempotent on document_id. Errors → Exception. No double post.

Not owned: RAG contents, intake doors, review UI, prompts.

Later adapters (do not staff now): Zoho Books, QBO, S/4 OData, B1 Service Layer, ECC BAPI, Fusion REST, EBS open interface.

Shared connector contract

Every adapter, including Tally, implements:

```
connect()
fetch_masters()
post_invoice(draft)
post_credit_debit(draft)
fetch_open_bills()
post_payment(draft) # optional, after invoice post is stable
health()
```

Posts are idempotent on document_id. Store erp_id + raw response on PostingJob.

Tally runtime

```

API / post worker
|
Person 4 site agent on client PC
|
localhost:9000 Tally Prime XML
|
company currently loaded in Tally

```

The ERP is not exposed to the public internet. Same agent class will later cover on-prem SAP/EBS.

Offline path: if Tally is not running, download the XML for the CA to import. Bill stays approved / post_pending, never silently dropped.

Mapping table

RAG maps to our ledgers first (Person 1). Person 4 then translates:

Our key	Tally
Vendor GSTIN	Ledger name (party)
Purchase / expense ledger	Ledger name
GST 18% intra	Input CGST + Input SGST
GST inter	Input IGST
TDS 194J	TDS ledger
Bill-wise	New Ref = invoice_no, due date

If the Tally vendor ledger does not exist: set create-proposal. Reviewer must approve. Never auto-create in MVP.

Post flow

```

approved VoucherDraft
-> router picks adapter from client.erp_type
-> adapter.validate(draft, masters)
    # company loaded, ledgers exist, period open-ish, tax ledgers present
-> adapter.post()
    Tally: XML purchase / DN / CN + bill-wise
-> store erp_id
-> attach source PDF (where Tally allows; else keep link in narration / extra file)
-> fetch open items for ageing
-> ERP errors -> exception
    NO retry that can double-post

```

Voucher types: purchase invoice, debit note, credit note, later payment with bill-wise refs.

Agent responsibilities

- Talk only to the loaded company.
- Health: Tally up, port 9000, company name.
- fetch_masters: company name, ledgers, groups, outstanding payables.
- Write masters snapshot to Person 2 tables so Person 1 can embed under client_id.
- Import purchase / DN / CN / payment.
- Create ledger only after reviewer approved “new vendor.”
- Idempotency: if PostingJob already has erp_id for this document_id, return that id. Do not import again.

XML rules

- Person 1 / Person 3 never generate XML.
- Templates live in the adapter. Draft fields fill slots.
- Bill-wise New Ref must equal invoice_no from the draft.
- Narration from draft, plus finledger:document_id= for support.
- Amounts from draft only. Do not re-compute GST in the adapter except to reject mismatch.

Interfaces

From Person 3

- approved VoucherDraft
- idempotency key
- attachment document_id

From Person 2

- encrypted `erp_link` row (agent pairing token, company name)
- object store for XML artifacts / request logs keyed by `client_id`
- `mapping_table` persistence

To Person 2 / Person 1

- masters snapshot after connect and on a schedule
- posted bill pointer so RAG can index last N invoices

To Person 3

- `PostingJob` success (`erp_id`) or exception (Tally error text, no retry storm)

Week plan (Person 4)

Week	Ship
0	Agent hello + company read + health
1	<code>fetch_masters</code> → empty mapping table
2	XML preview of draft (no post)
3	Post on approve + <code>erp_id</code> + attach / store PDF
4	Idempotent post, DN/CN, outstanding sync

Do not start Zoho or SAP in these four weeks.

Acceptance tests

- Health fails cleanly when Tally is closed.
- `fetch_masters` returns ledgers for the loaded company only.
- Preview XML contains party, purchase, GST lines, bill-wise ref = invoice_no.
- Approve twice → one Tally voucher.
- Missing party ledger → reject to Exception with create-proposal, no silent create.
- Tally rejection text lands on the ReviewTask exception tray.
- Masters snapshot is client-scoped.
- Offline XML file is produced when port 9000 is down.

Refusals for this seat

- LLM-written XML
- Auto-create vendor / ledger
- Fire-and-forget import
- Retry that can duplicate
- Exposing Tally to the public internet
- Building S/4 / Fusion adapters before the Tally loop is reliable
- Pretending ECC speaks S/4 OData when that work starts later